

ISHA RANA

Delhi, India | +91-9871012639 | rana.isha2016@gmail.com | LinkedIn | GitHub | LeetCode

PROFESSIONAL SUMMARY

Computer Science undergraduate (B.Tech) with basic experience in backend development, AI-based projects, and IoT systems. Familiar with building simple REST APIs and working with Python, databases, and related tools. Currently learning Data Structures & Algorithms and exploring system design concepts. Have worked on projects involving automation and real-world problem solving, with a growing interest in developing scalable and practical software solutions.

TECHNICAL SKILLS

Programming Languages: Java (Intermediate), Python (Basic), SQL

Backend & Frameworks: Java Spring Boot, REST API Development, Microservices Architecture

AI & Machine Learning: OpenAI API, Large Language Model (LLM) Integration, Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP)

Databases: PostgreSQL, MySQL, MongoDB

Tools & Testing: Git, Postman, Swagger / OpenAPI, IntelliJ IDEA, VS Code

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design, Problem Solving

PROJECTS

AI-Driven Automated Beverage Dispensing & Billing System *Feb2026 – Present*

Python (FastAPI), MySQL/PostgreSQL, HTML/CSS/JavaScript (or Streamlit), AWS/Azure Cloud, ESP32 (IoT), Pandas/NumPy (AI), REST APIs, Git, Postman

- Developing an **AI-driven automated beverage dispensing and billing system** where **payment is held and finalized only after accurate dispensing**, ensuring transactional atomicity.
- Building a **real-time dispenser simulator** to stream flow data and enable hardware-free testing of dispensing, billing, and system logic.
- Implementing **volume-based auto billing** with a reconciliation layer to match poured quantity with billed amount, improving auditability and traceability.
- Designing a **fraud detection module** to identify over-pouring, under-pouring, and abnormal patterns, along with a dashboard for real-time monitoring and insights.

Skill-Based Job Recommendation and Alert System *JAN2026-APR2026*

Python, FastAPI/Flask, MySQL, REST APIs/Web Scraping, SMTP (Email Automation), HTML/CSS/JavaScript

- Designed and developed a skill-based job recommendation platform that matches user profiles (skills, roles, location) with newly posted job opportunities.
- Implemented automated job fetching (APIs/web scraping) and intelligent filtering to deliver personalized recommendations with direct application links.
- Built a daily email notification system using SMTP to send curated job alerts, reducing manual job search effort.

EDUCATION

Bachelor of Technology — Computer Science

2024 – 2028

GLA University, Mathura, Uttar Pradesh | CPI: 7.6 / 10.0

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, System Design

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- Achieved a **Top 10 position** in **UI/UX Design and Technology** at **Imagix, Galgotias University**.
- Achieved **OCI Data Science Professional Certification (Oracle)**, demonstrating expertise in ML model building and deployment.

AREAS OF INTEREST

Backend Development • System Design • LLM & AI Integration • Data Structures & Algorithms • Cloud Computing (AWS) • Microservices Architecture • Open Source Contribution